

# Designing for Ecological Needs and Pervasive Information Architecture

## Highlights

- Designing for ecological needs
  - Foundational layer of the user needs pyramid
  - Ecology is about usage context
  - Information structures must pervade the ecology
  - Ecological design spans interaction channels
  - For the user, the entire ecology is a single service
- Creating an ecological design
  - Example: An ecology for a shopping application within a pervasive information architecture

## 15.1 INTRODUCTION

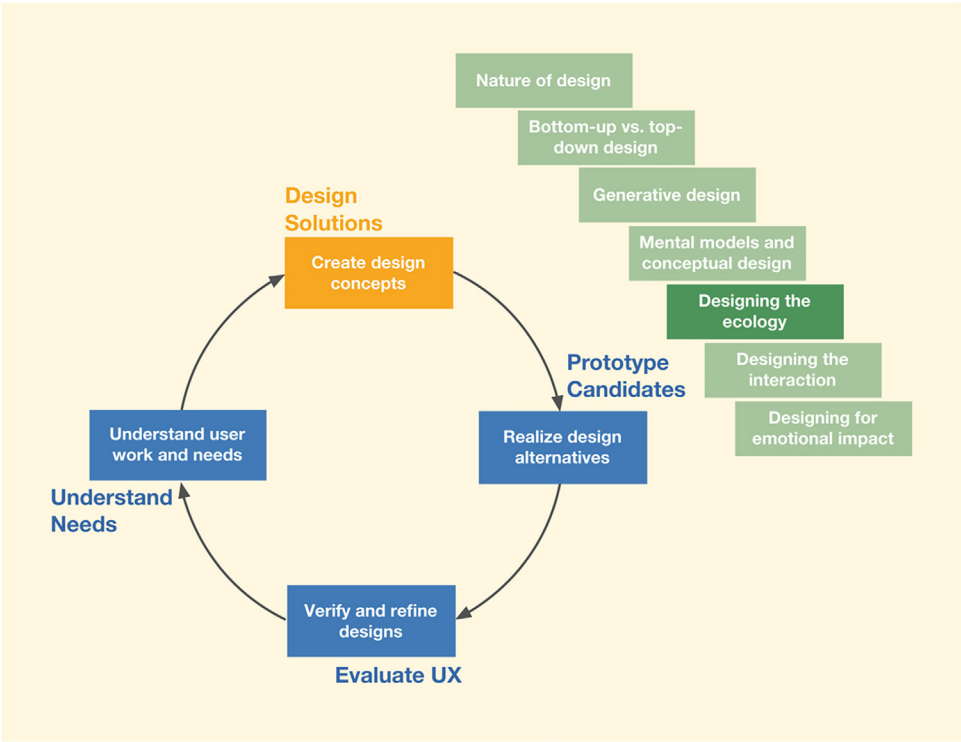
### 15.1.1 You Are Here

We begin each process chapter with a “You are here” picture of the chapter topic in the context of The Wheel, the overall UX design lifecycle template (Fig. 15.1). In Section 11.2.1, we discussed how ecological needs are about the overarching and encompassing requirements, constraints, and activities of the work practice beyond just the product or system being designed. In this chapter, we go about designing the ecology to satisfy those needs of users with special attention to information architecture that is pervasive across the ecology.

## 15.2 DESIGNING FOR ECOLOGICAL NEEDS

### 15.2.1 Ecological Design: Foundational Layer of the Needs Pyramid

In the setting of UX design, the ecology is an entire set of surrounding parts of the world, including networks, other users, devices, and information structures with



*Fig. 15.1*  
*You are here in the chapter describing designing the ecology within the design solutions activity, in the context of the overall Wheel UX lifecycle.*

which a user, product, or system interacts. Designing for ecological needs is one of the most overlooked aspects in UX practice. The lure to get a sense of progress by diving right into interaction design (screen and visual designs) is often strong because it is easier to understand the tactical needs of the user at a task level and translate them into corresponding widgets or patterns in the user interface.

For users to be productive in the work practice, however, they need to first understand the broader ecology and to participate in it. Designing the ecology forces us to think about the overall system and how it addresses broader work activities.

**Example: Apple Music and Satisfying Ecological Needs First**

Consider the work practice of listening to music. Here we describe how users interact with the Apple ecology through the Apple Music application. If users are handed a new device, such as an iPhone, and asked to listen to a favorite song stored on their personal computer (Mac), they will not be able to do that directly without first understanding how the Apple ecology works and doing some activities to set it up.

Understanding the ecology entails users to determine how to participate in Apple’s ecosystem by registering an iCloud account and setting up an iTunes

Match subscription in the Apple Music application to import, organize, classify, manipulate, and listen to music stored on their computer, plus syncing their music library to their other devices, such as an iPad.

If users do not have their favorite song available on a Mac, then they need to browse or search for it on Apple's or some other compatible music store, then buy and download or stream that piece of music. Interacting with the iPhone to listen to music only happens after these ecological needs are satisfied.

In this example, the ecological needs are complex and can be a significant barrier to playing the music. That is the nature of this work activity, however, because of the various ecological requirements and constraints: legal (music industry's antipiracy requirements), platform (Apple's ecosystem requiring users to deal with iTunes Match, Apple ID, Apple Music), device (iPhone's wi-fi or cellular capabilities), user (existing music library saved in a device vs. subscriptions and setting up playlists on streaming services).

### 15.2.2 Designing the Ecology Is About Usage Context

Designing the ecology is about envisioning and planning how work gets done in the broadest context of the user and system. This means looking at the following:

- Devices users employ to get work done
- Variations in their form factors and capabilities (e.g., watches, phones, tablets, laptops, desktops, wall displays, ambient sensors)
- Other products and systems that are part of the work practice
- Usage contexts (e.g., usage while sitting at a desk vs. on the move)
- Infrastructure constraints (e.g., with connectivity vs. without)

### 15.2.3 Information Architecture

We cannot discuss the nature of design without at least mentioning information architecture. These concepts of information architecture are deeply intertwined and offer overlapping perspectives originating from different fields of study. This topic is much too broad to cover here, but we include a mention of it for completeness.

#### 15.2.3.1 Information architecture is design

In the context of design, information is much more than just data, which is about facts and figures (Morville & Rosenfeld, 2006) and databases that are highly structured collections of data. Information is about how knowledge is represented, which is more useful to most users. Information architecture

is about designing information for effective usage, including organizing, structuring, and labeling information. It involves methods for storing, searching for, and retrieving information.

Much of the challenge of information architecture in UX design is wrapped up in finding ways to support searching and findability. It is also about presenting information in ways that users can understand and that help people make use of information (e.g., in making decisions on a website).

Within the context of design, information science is a sibling concern of UX, focused on “the analysis, collection, classification, manipulation, storage, retrieval, movement, dissemination, and protection of information” (Stock & Stock, 2015).

A related area that owes its roots to information science and the field of architecture is information architecture. Wurman (1997), an architect by training and a graphic designer by choice, is regarded to be the inventor of “information architecture”.

Morville and Rosenfeld (2006) define information architecture as the design of information structures for organizing, storing, retrieving, displaying, manipulating, and sharing. Information architecture also overlaps the very large UX areas of design for labeling, searching, and navigating information.

### 15.2.3.2 *Pervasive information architecture*

A pervasive information architecture is a structure for organizing, storing, retrieving, displaying, manipulating, and sharing information that provides ever-present information availability spanning parts of a broad ecology (Resmini & Rosati, 2011).

UX design for the ecology almost always depends on pervasive information architecture and the flow of information across system or device boundaries within a greater ecology. This allows users to interact with the same information, perhaps in different forms and accessed and displayed in different ways, on different devices at different times and in different places.

### 15.2.4 *Ecological Design Spans Interaction Channels*

An interaction channel is a means, mode, or medium through which users and parts of a system interact and communicate, including sensory modes, such as visual communication, as well as voice and tactile interaction. The concept also includes such devices as desktop computers, smartphones, and system-oriented channels, such as the internet, wi-fi connections, and Bluetooth.

In the information architecture community, ecological design is sometimes referred to as cross-channel information design or pervasive information design (Resmini & Rosati, 2011). In the human–computer interaction community, ecological design is sometimes called multi-platform user interface (Pyla, Tungare, & Pérez-Quñones, 2006) or continuous user interface (Pyla et al., 2009).

When working at a computer, we are usually doing single-platform computing. When our work also includes tablets or smartphones, it extends to multi-platform computing.

In an ecology, a single service and the associated pervasive information architecture can be spread across multiple platforms, all of which are required to make it work (Houben et al., 2017). As an example, your bank sends a text message confirming a transaction you started online but came to the bank to finish. Pervasive information architecture is required to support ubiquitous computing (Weiser, 1991), technology that resides almost anywhere in a user's ecology, including in appliances, homes, offices, stereos and entertainment systems, vehicles, roads, and objects they carry (e.g., briefcases, purses, wallets, wrist watches).

### 15.2.5 For the User, the Entire Ecology Is a Single Service

Consider this example of how an extensive ecology provides a single integrated user service. Suppose a user orders something online using a computer and the internet, opting for free shipping to the local store. Shortly thereafter, the user gets an automated email with a purchase confirmation and a copy of the invoice. Sometime later, the customer gets an email saying that the item has been shipped, giving a tracking number. The user clicks on this tracking number a few times in the next day or two and sees the progress of the shipment.

An email with a bar code and a text message later announce the arrival of the item at the store, and soon the person goes to the store physically to pick it up. The user takes a copy of the email on a smartphone to the store, and the cashier scans the barcode. The user signs a tablet at the store, confirming pickup. Sometimes another email confirms the pickup and closes out the transaction. Lots of different interactions, platforms, and devices are involved, all serving one user transaction.

Although we may think of goods and services as separate things, Norman (2009) shows how the real value in a product or service is to offer a great experience to the customer/user by calling us to systems thinking (in the form of ecological design) to provide that value to users.

## Example: A Flawed Ecological Design Can Cause User Frustration

A good ecological design empowers users to perform whole activities in the work domain (high-level support to work activities), not specific workflows or tasks. A flawed ecological design results in inconsistent experiences for users as they navigate the work domain in the ecology. Consider this issue we recently encountered that illustrates the difference between thinking about ecology and thinking about interaction, in the context of Apple's ecological design.

A user's ecology in the Apple world contains desktops, laptops, tablets, phones, and smartwatches. Depending on the context of interaction, the user may use any one of these devices to access information. When Rex wanted to look for some details from a previous doctor's appointment on his iPhone, for example, the search came up with no results. He was surprised because he knew all his annual appointments for medical checkups are tracked in his calendar application. As a check, he tried the calendar on his desktop and, surprise, it was there.

The cause of this mismatch of device behavior was a default setting on his iPhone by which only appointments from the last month are saved, and his last appointment fell outside of that window. There was no indication on the search results page that this was the case; this constraint on the search results is hidden somewhere many levels deep in the device settings. On the desktop calendar application, however, there is no such restriction.

The problem is not with that ecological design choice but with the way designers failed to reflect it or explain it. It appears the designers responsible for the iPhone search capability focused only on the interaction design on the phone (how search results are handled and the default calendar scope). From the user's perspective, however, the phone should not be different from the desktop when it comes to calendar information. It was supposed to be just a different device in the same ecology; for the user, complete calendar information should have spanned that ecosystem.

As an aside, we point out that this is a case where the user's mental model failed to align with the designer's and where a better conceptual design could have alleviated this breakdown that Rex experienced.

As we talk about designing for the ecology here, this small example clearly illustrates the difference between designing for the ecology and designing only for the interaction. If Apple designers had thought of the ecological design, then they would have supported the search task more effectively while still considering the constraints of the smartphone

device. Perhaps they could have offered an option along the lines of “No search results found in the last 30 days. Continue the search on the server (iCloud)?” or something similar.

## 15.3 STORYBOARDS

### 15.3.1 What Is a Storyboard?

Storyboards bring a design to life in graphical clips (i.e., freeze-frame sketches illustrating stories of how people will work with the system). A storyboard is a sequence of these sketches as frames illustrating the interplay between a user and an envisioned ecology or device. This narrative description can come in many forms and at different levels of detail.

Storyboards for representing interaction sequence designs are like visual scenario sketches, depicting envisioned interaction design solutions. A storyboard may be thought of as a comic book–style illustration of a scenario, with actors, screens, interaction, and dialogue showing sequences of flow from frame to frame.

### 15.3.2 Storyboards Can Cover All Layers of the Pyramid

Even though storyboards are predominantly about how users interact with the envisioned system, they can cover aspects of the other two layers, too. Users can be depicted at all touch points they have with the ecology. Begin by creating illustrated sequences that show users interacting with the system in a narrative style.

Include things such as the following in your storyboards:

- Hand-sketched pictures annotated with a few words
- All the work practice that is part of the task, not just interaction with the system or ecology (e.g., phone conversations with agents or roles outside the system)
- Sketches of devices and screens
- Sketches of the user’s journey within the system ecology
- Any connections with system internals (e.g., flow to and from a database)
- Physical user actions
- Cognitive user actions in thought balloons
- Sketches to convey emotional impact

Because storyboards illustrate how users go about meeting their goals while envisioning the high-level interplay among human users, the different devices in the ecology, and the surrounding context, they are valuable

when demonstrating the potential of the system in a context where it solves particular problems. To do this, you could show a device in the hands of a user and connect its usage to the context. As an example, you could show how a handheld device is used while waiting for a flight in an airport.

Storyboards can also focus exclusively on interaction designs (see [Chapter 16](#)) and show screens, user actions, transitions, and user reactions for each device. You may still show the user, but now it is in the context of user thoughts, intentions, and actions upon user interface objects in operating the device. Here is where you get down to concrete task details. Select key tasks from the hierarchical task inventory (see [Section 9.2](#)), design scenarios (see [Section 9.3.1](#)), and task-sequence models (see [Section 9.3](#)) to feature in your interaction perspective storyboards.

Storyboards can also be used to envision emotional design aspects and to illustrate deeper user experience phenomena, such as fun, joy, and aesthetics. They can show the experience itself—remember the excitement of actually riding the mountain bike in the example from Buxton (2007b) (see [Section 1.4.3](#)).

### 15.3.3 Storyboards Help Deflect Focus From Technology to Interaction

Sometimes you may find yourself frustrated with a project when the team is focusing on backend technology and not interaction with users or customers. Starting your interaction designs with storyboards before going to wireframes and prototypes is an effective way of redirecting that attention to designing the ecology or interaction. This allows the team to start by visualizing and refining the interaction with the ecology or a device to be designed into wireframes.

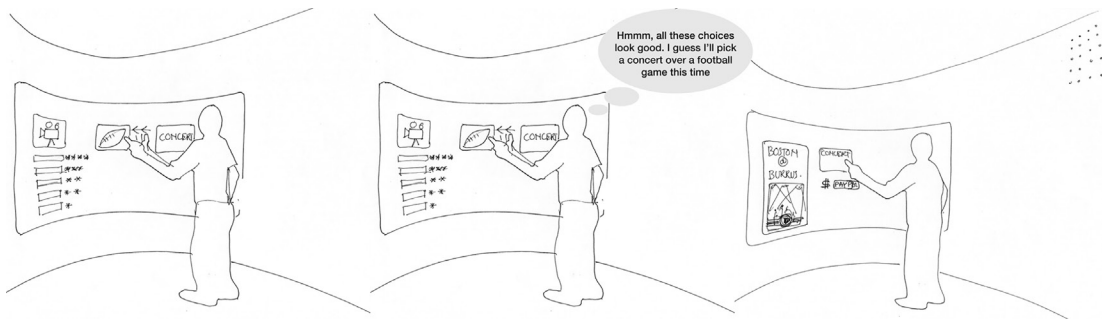
### 15.3.4 Importance of Between-Frame Transitions

Each storyboard frame shows an individual state as a static screenshot. Through a series of such snapshots, storyboards show the progression of interaction over time. The spaces between the frames are where the transitions are made, and these transitions are where the user experience lives (Buxton, 2007b). Therefore the actions between frames should be part of the sketch. The transitions are where the cognitive affordances in your design earn their keep, where most problems for users exist, and where the challenges lie for designers.

We can augment the value of our storyboards greatly to inform design by showing the circumstances that lead to and cause the transitions and the context, situation, or location of those actions. These include user thoughts, words, gestures, reactions, expressions, and other experiential aspects of

interaction. Is the screen difficult to see? Is the user too busy with other things to pay attention to the screen? Does a phone call lead to a different interaction sequence?

Fig. 15.2 shows a transition frame with a user thought bubble explaining the change between the two adjacent state frames. This thought bubble says, “Hmmm, all these choices look good. I guess I’ll pick a concert over a football game this time.”



*Fig. 15.2*  
Storyboard transition  
frame with thought bubble  
explaining state change  
(from Akshay Sharma,  
Virginia Tech Department of  
Industrial Design.).

### 15.3.5 Storyboards for the Ecological Perspective

Storyboards are particularly useful in describing all the ways the system ecology overlaps with the user’s work context and how it supports (or not) their needs. Storyboards can clarify where technology can intervene by highlighting the opportunities the user’s work context affords for designers to leverage.

#### Example: Storyboard Sketches for the Ticket Kiosk System Ecology

Fig. 15.3 provides an example of a sequence of sketches as a storyboard depicting specific use of the Ticket Kiosk System. This storyboard depicts how a kiosk at a bus stop (part of the kiosk ecology) can facilitate opportunistic interactions with users as they wait for the bus. Users have time to think while waiting for the bus. Because the kiosk is part of the bus stop, it provides a good opportunity to get movie tickets while waiting.

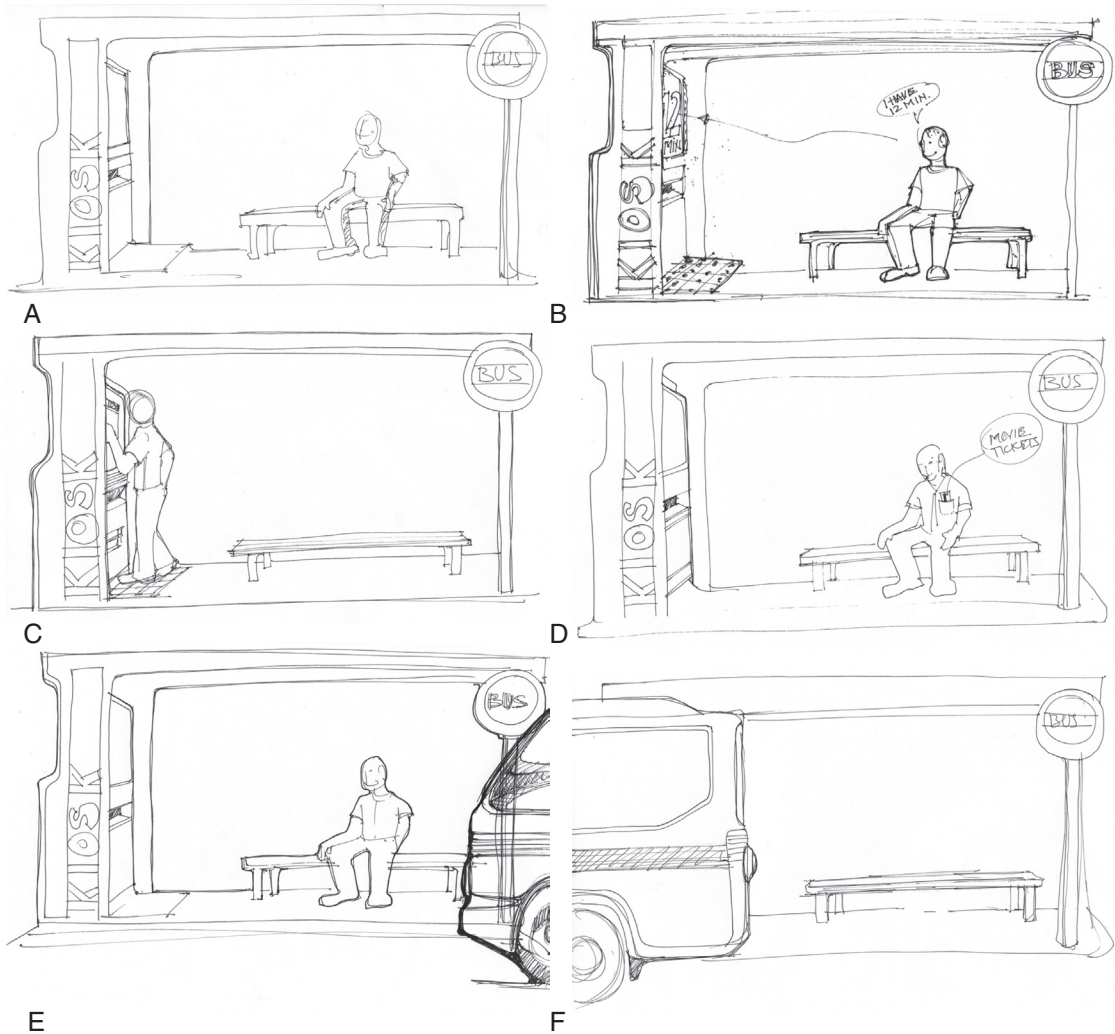


Fig. 15.3

Example of a sequence of sketches as a storyboard (from Akshay Sharma, Virginia Tech Department of Industrial Design.).

### Example: More Ticket Kiosk System Storyboard Sketches for the Ecological View

Fig. 15.4 continues with part of a different Ticket Kiosk System storyboard. This storyboard depicts a similar interaction in a shopping mall context, and how a kiosk can potentially trigger a planning activity with friends involving entertainment events.

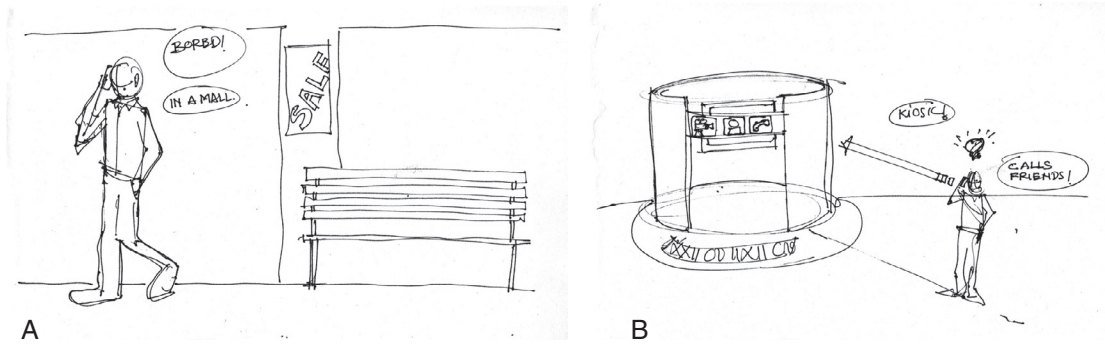


Fig. 15.4

Part of a different Ticket Kiosk System storyboard (from Akshay Sharma, Virginia Tech Department of Industrial Design.).

## Exercise 15.1 Storyboard for Your Product or System

**Goal:** Practice sketching storyboards (see Section 5.6.1.2).

**Activities:** Sketch storyboard frames illustrating narrative sequences of action in the ecological perspective for your product or system. Illustrate high-level interplay among human users, the system as a whole, and the surrounding context.

**Schedule:** You decide how much time you can afford to give this.

## 15.4 CREATING AN ECOLOGICAL DESIGN

### 15.4.1 Representing Ecological Design

An ecology's design is usually represented using a concept diagram that looks like a flow model and identifies the following:

- System entities
- Work roles
- Propagated information
- User tasks
- Ecology boundaries
- External dependencies
- Description of the underlying metaphor or theme of the concept (e.g., a central mothership-like service coordinating a set of connected mobile devices)
- How they all fit together

### 15.4.2 Identify Sub-systems by Related Work Roles

In [Section 8.2.1](#), we talked about how the workflow of certain work roles has little or no overlap with those of others. In many systems, these parts of the overall design are almost mutually exclusive, allowing for a logical partitioning of the target system.

For example, in the Middleburg University Ticket Transaction Service (MUTTS) work practice, the nature of work done by a ticket buyer is different from that of the event manager. They rarely, if ever, interact with each other within the ecology of MUTTS. Each work role can be thought of as associated with a sub-system or an ecology in itself. This means the overall ecology of the work practice can be thought of as having a set of ecologies, one for each sub-system. This separation provides us a way to reduce overall design complexity through a divide-and-conquer approach.

The sub-systems themselves are connected via information exchange (e.g., through a central database) and specialized work roles responsible for that exchange at their boundaries. For example, there are work roles, such as systems or IT engineers, responsible for ensuring that the ticket information from event vendors is correctly stored in a database that is available accurately and reliably to ticket buyers.

### 15.4.3 Proceed With Generative Design

Using ideation and sketching techniques we discussed in [Chapter 13](#), synthesize ideas for what the ecology is, what it contains (including devices and channels, features, capabilities), and most importantly a conceptual design to explain how it works. The goal is to create as many ideas as possible for the overall theme or metaphor of the ecological conceptual design. After the ideas are generated and captured as sketches, the team will critique each idea and concept for tradeoffs.

### 15.4.4 Establish a Conceptual Design for the Ecology

The ecology may feature a mothership-like concept, in which there is a central entity that acts as the brain and information exchange for all other entities and devices in the ecology. Or, the ecology may be better served by a peer-to-peer architecture without a central repository concept. How will such a concept address the breakdowns, constraints, or other deficiencies in the work practice? As part of this phase, you should also come up with ideas on the pervasive information architecture.

### 15.4.5 The Issue of Self-Sufficiency

An ecology's self-sufficiency comes down to how the ecology is structured and whether it provides everything the user needs to thrive in the ecology without depending on other ecologies.

As an example, Norman (2009) cites the Amazon Kindle—a good example of a self-sufficient ecology. The product is for reading books, magazines, or any textual material. You do not need a computer to download or use it; the device can live as its own independent ecology. Browsing, buying, downloading books, and more is a pleasurable flow of activity. The Kindle is mobile, self-sufficient, and works synergistically with an existing Amazon account to keep track of the books you have bought through [Amazon.com](http://Amazon.com). It connects to its ecology through the internet for downloading and sharing books and other documents. Each Kindle has its own email address so that you and others can send lots of materials in lots of formats to it for later reading.

### Example: Conceptual Design for the Ticket Kiosk System Ecology

For the Ticket Kiosk System, we envision a central control system that acts as a hub for all transactions. This is necessary to keep track of ticket inventory and for managing temporary locking of tickets while users are in the middle of a buying transaction. If the transaction was abandoned or times out, those tickets will be released for other customers. One downside of this concept is that the central hub can become a single point of failure, where breakdown in the hub can bring down all kiosks and websites.

[Fig. 15.5](#) provides an early conceptual design sketch for the Ticket Kiosk System from the ecological perspective. All transactions from all kiosks and the website where tickets are sold are served by the control center. Users bring their own personal ecologies into the overall flow when they use laptops, tablets, and phones to search, browse, and buy tickets. At least some of these user devices can be used to contain e-tickets in lieu of physical tickets. This means that each ticket must have an ID number that can be scanned by the Ticket Kiosk System devices at event venues.

The team came up with other ideas, such as smart tickets that were aware of a venue's layout ([Fig. 15.6](#)). During the critiquing phase, issues arose about feasibility with available technology. The idea was then modified to add, at key locations near the entrance, a new feature of the ecology: a ticket scanner that shows a path to the corresponding seat location.

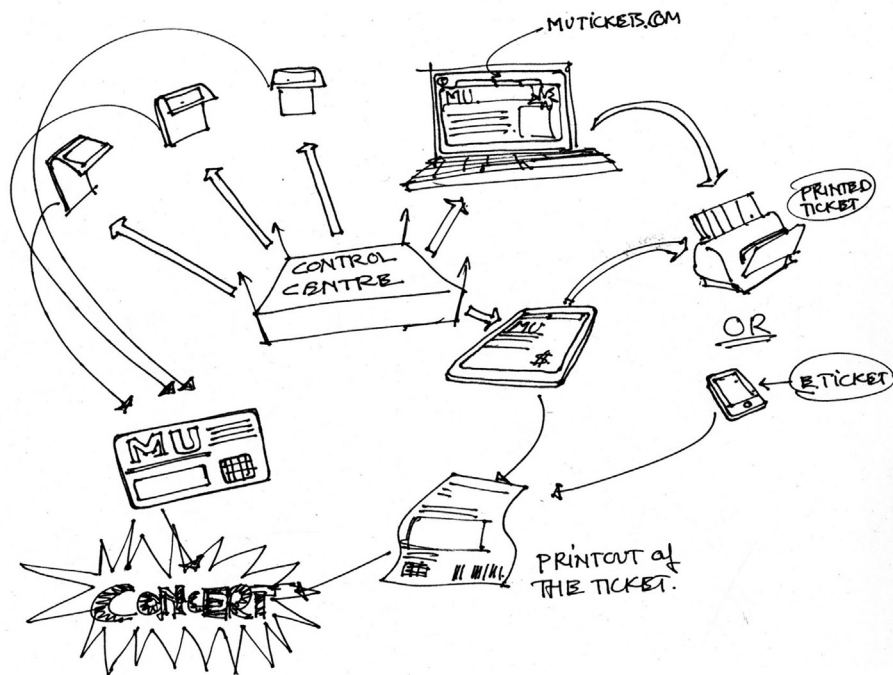


Fig. 15.5

An early conceptual design idea from the ecological perspective (from Akshay Sharma, Virginia Tech Department of Industrial Design.).

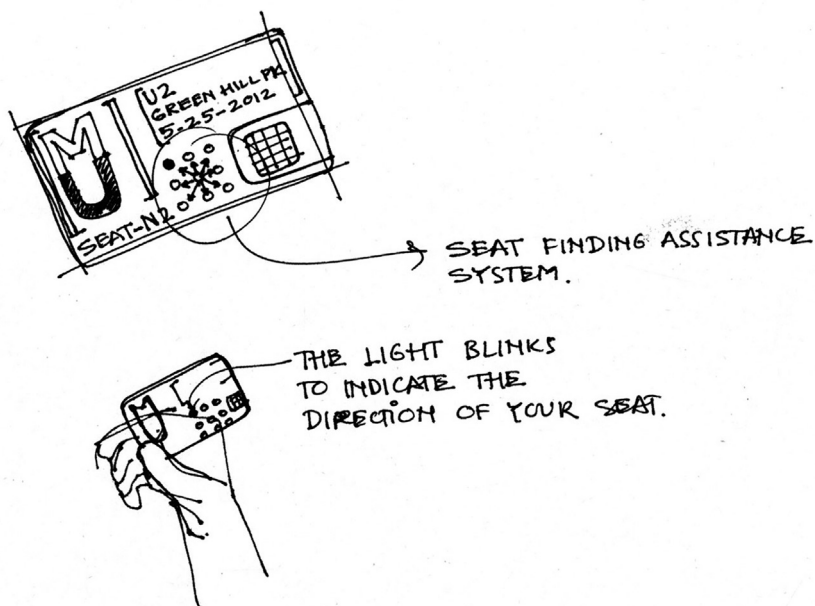
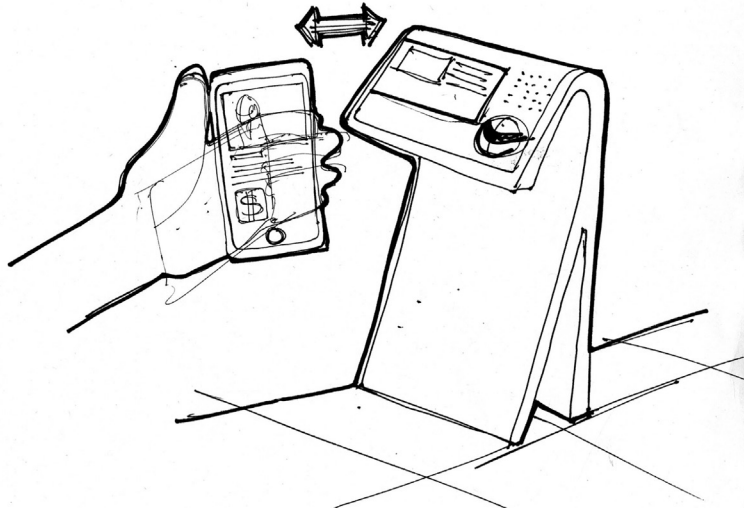


Fig. 15.6

Ecological conceptual design idea focusing on a feature for a smart ticket to guide users to seating (from Akshay Sharma, Virginia Tech Department of Industrial Design.).

Fig. 15.7 provides an ecological conceptual design idea for the Ticket Kiosk System focusing on a feature showing communication connection with a smartphone. You can have a virtual ticket sent from a kiosk to your mobile device and use that to enter the event.



*Fig. 15.7  
Ecological conceptual design  
idea focusing on a feature  
showing communication  
with a smartphone (from  
Akshay Sharma, Virginia  
Tech Department of  
Industrial Design.).*

Fig. 15.8 provides an ecological conceptual design idea for the Ticket Kiosk System focusing on the features for communicating and social networking to share information about event attendance.

This is a good example of the part a full ecology can play in the life of the user. Storyboards (Section 15.3) are also an excellent tool for capturing this part of the ecological perspective, the context of use.

## Exercise 15.2 Conceptual Design for the Ecology of Your Product or System

Think about your system and contextual (UX research) data and envision a conceptual design, including any metaphors, in the ecological perspective (see Section 5.6.1.2). Try to communicate the designer's mental model, or a design vision, of how the system works as a black box within its environment.

## 15.5 DESIGNING AN ECOLOGY TO INFLUENCE USER BEHAVIOR

Beale (2007) introduced an interesting concept he called “slanty design,” which is used to discourage certain undesirable user behaviors by selectively

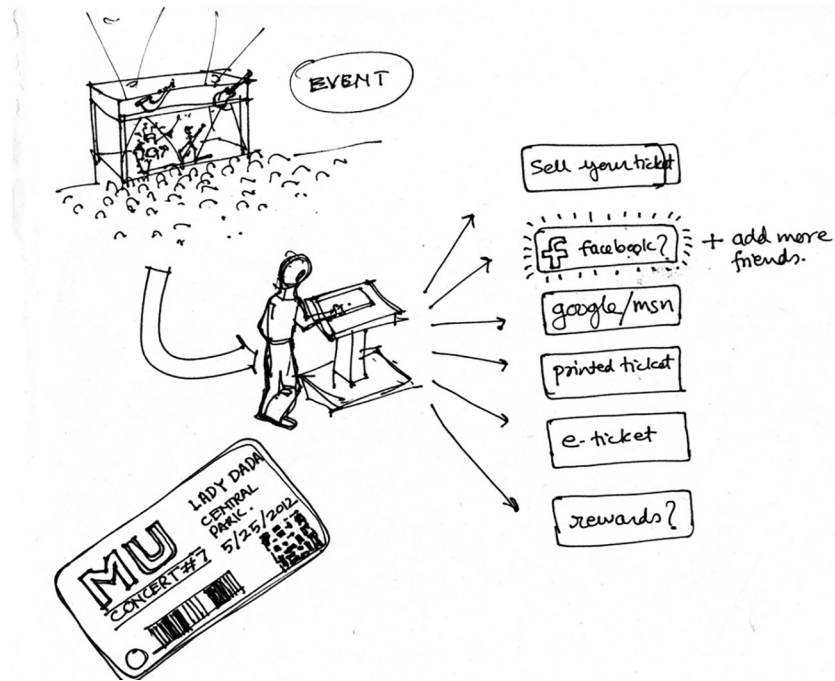


Fig. 15.8

Ecological conceptual design idea focusing on the features for communicating and social networking (from Akshay Sharma, Virginia Tech Department of Industrial Design.).

reducing functionality and/or usability, making it difficult to do things not in the interest of other users or the enterprise in the ecological perspective, but still allowing individual users to accomplish the necessary basic functionality. A simple example is a conical cup that users are unlikely to set down on a table.

Another example is sloped reading desks in a library, which still allow reading but make it difficult to place food or drink on the desk or, worse, on the documents. Beale (2007) provided a similar example in the domain of airport baggage claims. People naturally gather around the baggage conveyor belt, and many people even bring their carts with them. This behavior increases usability of the system for them because the best ease of use occurs when you can just pluck the baggage from the belt directly onto the cart.

However, crowds of people and carts cause congestion, reducing the accessibility and usability of other users with similar needs. Signs politely requesting users to remain away from the belt except at the moment of luggage retrieval are regrettably ineffective. Beale (2007) proposed a slanty design for the baggage carousel to solve this problem. In this case, it involves something that is physically slanted; the surrounding floor slopes away from the baggage carousel.

This interferes with bringing carts close to the belt and significantly reduces the comfort of people standing near the belt by forcing them to remain away from the carousel until they can dash for the bags when they arrive within grasping distance. This reduces one aspect of individual usability somewhat, but he claims that it works best overall for everyone in the ecological perspective. Slanty design should include evaluation to eliminate unforeseen and unwanted side effects, such as people slipping or falling while navigating the slope to the carousel.

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## 15.6 AN ECOLOGY FOR A SMART SHOPPING APPLICATION

We conclude this chapter with an extended example of designing the ecology for a shopping application based on pervasive information architecture, which we adapted from Resmini and Rosati (2011). This is also a good example of an activity-based design (see [Section 1.2.4](#)), which is about interaction in the context of one or more task thread(s), a set of (or possibly sequences of) multiple, overlapping, and related tasks, often involving more than one device in an ecology.

### 15.6.1 High-Level Issues

**Who is the client?** While we are designing for users to have a good shopping experience, our client is the store management. You cannot make an app to help customers unless the store participates in the ecology, which cannot occur unless store management is the client.

**It is hard work to find things in a store.** Even after shopping in the same large store dozens of times, many shoppers do not feel they know where things are, and there is seldom enough staff on hand to assist. As a result of just this one factor, the experience of shopping in a large physical retail store can be frustrating and exhausting.

**The dilemma of impulse buying.** The UX team came up with a design idea that makes it easy for customers to locate products within the store, an efficient customer shopping experience that should be welcomed by all. Typical store management, however, does not want shoppers to know where everything is—at least not immediately—without having to browse through the store to find it. As you may know, many large stores design their merchandise layouts to slow shoppers down, exposing them to impulse buying opportunities—displays of items they had no plan on buying, but may find interesting enough to buy anyway.

The layout of some stores is deliberately changed periodically to offset shoppers learning where things are. Even experienced shoppers are now forced to browse large portions of the store layout while trying to avoid the impulse item traps.

How can designers support the user need for efficiency in the face of this behavior by store management? The answer goes back to basics—thorough UX research. During UX research, you have to explore the store perspective of the shopping activity; through UX research, your team may recognize that impulse buying is not always bad for the shopper. Customers may occasionally buy one of these items that is really nice, useful, or even fun—good for both customer and store. So, you need a design to sell store management on streamlining the user shopping experience while still retaining impulse sales. Your design challenge is to:

- Replace the usual frustrating and exhausting shopping experience with one that is efficient and actually fun
- Convince store management that this alone can bring customers back more often, offsetting any lost impulse sales
- Find other ways to include impulse shopping

## 15.6.2 Key Parts of the Design

**Smart fridge.** We will be designing a shopping experience for a customer that starts with a smart fridge in the home, a fridge that knows about its contents and learns about user needs and food preferences. It can, for example, tell when the milk supply is dwindling and suggest putting milk on the shopping list. The ecology can include a smart pantry that notifies when the last item of a type is removed from its shelves.

**SmartShop mobile app.** We will assume that users who are savvy enough to be using a smart fridge will almost always carry a mobile device, such as a smartphone. As part of our ecology, we will design a smartphone app called SmartShop.

**SmartKart.** A key part of this project is the design of SmartKart, which helps users find things in the store. SmartKart will contain a specialized built-in mobile device with a touchscreen display, connected to the store ecology via a special wi-fi setup. The SmartKart processor can also be used offline for mundane tasks, such as doing price/unit cost comparisons.

## 15.6.3 How It Works

**Pre-shopping activities.** The following user activities must occur before the shopping can begin:

1. Accumulate a generic list of things to buy (e.g., soup).
2. Match the generic items to specific products in the store (e.g., Campbell's tomato soup concentrate, 8 oz. can):
  - a. The SmartShop app finds candidate matches.
  - b. The user mediates by selecting the best match among candidates.
3. The SmartShop app generates a specific shopping list.
4. At the store, the user downloads the specific shopping list to SmartKart.

**Accumulating a generic list.** Step 1 of the process is the accumulation over time of a generic shopping list, expressed with as much or as little detail and specificity as desired. An example of a generic item is orange juice. We will make it easy for users to grab the phone to do step 1, entering a new item into the current generic list in SmartShop when they think of something to buy (left-most part of Fig. 15.9).

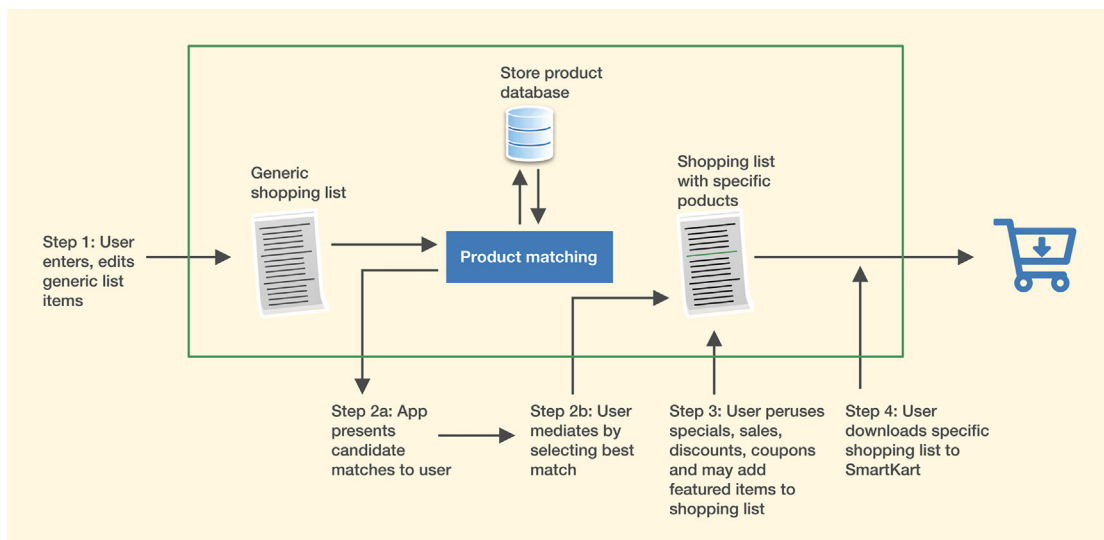


Fig. 15.9

Pre-shopping trip activities.

**Matching generic list items to specific store products.** When the user is ready to make a shopping trip, the focus moves to step 2, where SmartShop is used to match each generic list item to a specific product known to be in the store (lower left in Fig. 15.9) by:

- Step 2a: SmartShop searches for a generic item using inexact database matching, leading to a set of specific candidate matches.
- Step 2b: User selects desired specific item from possible matches in search results.

For example, for a generic item of orange juice, a specific product may be “UPC code = xxxxx, Green Valley brand, fresh (not from concentrate), one quart, with pulp.”

**Generate a specific shopping list.** As each specific product is selected, in step 3 it is added to the accumulating specific shopping list in the app (bottom next to far right in [Fig. 15.9](#)).

Through UX research we also learned that users occasionally look through online ads and emails from stores about sales, specials, discounts, or coupons. These specials and discounts, if available, are shown to the user when presenting the specific items that match the generic entries. Clicking on the “Add to my shopping list” button while viewing one of these items will add that item to the specific list in the app.

**Download the specific shopping list to SmartKart at the store.** Upon arrival at the store, the shoppers perform step 4 by launching the SmartShop app on a mobile phone and tapping the phone on a shopping cart. This establishes a connection to the built-in mobile device of the SmartKart, which automatically syncs to the current specific shopping list (extreme bottom right in [Fig. 15.9](#)), making the specific shopping list pervasive in the ecology.

### 15.6.3.1 Finding things in the store

If a design goal is to help customers locate items efficiently, where is the information needed to make that happen? This is a good example of embodied information. Product-location information is embodied in the physical ecology of the store, including the items themselves. Each item could be aware of its location in terms of aisle, section, shelf, and so on. Conversely, each shelf could know what items are located there. Any number of existing technologies can capture this location information in a database.

**In-store location awareness of products on the shelf.** SmartKart browsing (sensing products as you pass them in the aisles) could be enough to help the user find things in the store. We can do better, however, if somehow the shelves themselves are embodied physical devices that know something about the products they contain, perhaps by something involving radio frequency identification sensors and/or bar code scanners built into the shelves. Awareness by the shelves of their contents will provide us with the necessary accurate product-location information, even if there are

errors in stocking. In addition, of course, the shelves must now be able to communicate (via the store wi-fi) what they know to carts and to the store database system.

Shelf-awareness of products offers a bonus: a continuous capability for keeping inventory information accurately up to date, eliminating the expensive, error-prone, inconvenient, and labor-intensive hassle of periodic manual store inventories.

Another potential benefit to the store is, as Resmini and Rosati (2011, p. 217) say in their grocery store example, that a product-location system will be liberating to the store. First, sales associates on the floor no longer have to help customers find things. Also, if the layout no longer has to be constrained by how customers find things in the store, it opens the door to innovative and dramatically different store layouts (e.g., layouts that sacrifice logical organization for the sake of efficient use of shelf space).

With these critical ecological design innovations, the act of moving the cart through the store can be mapped to moving the cart through items on the shopping list—almost.

***Cart awareness of its location in store.*** The next capability we need is obvious: The cart has to know where it is within the store. This could be accomplished by shelf-to-cart wireless communication, but perhaps a more flexible solution is a capability within the cart processor somewhat analogous to a localized GPS system and an internal map of the store. The in-store location system would not be a real GPS, however, but will be implemented via some kind of network with location beacon devices strung (perhaps in the ceiling) along each aisle. This process introduces into the store information architecture a key new information object: the current location of each cart.

***Cart ability to find a product.*** Now we have what we need for matching cart location to the location of products on the shopping list. In our pseudo-GPS app, the screen on the SmartKart shows a map of the internal layout of the store, showing aisles as roads on which to drive the cart, so to speak. The GPS display shows where the cart is within the store, and it knows the so-called addresses of each item on the shopping list. The GPS will treat each item on your shopping list as a destination, dropping pins for your items on the store map and ordering the list so that all items can be found efficiently in a single pass through aisles of the store.

As the cart approaches a store location containing an item on the shopping list, the cart display can sound an optional muted tone or flash the screen and indicate where to look to get the item. As the item is put in the cart, it will be checked off the shopping list, and a running total of the cost can be updated

using cost information obtained via the UPC bar code. All of this is the perfect cross-channel mapping from one device to another within the ecology that connects customer needs to store services.

This scenario is in the process of becoming a reality. Amazon has been evaluating a grocery store technology in which the system logs items as they are put in the cart so shoppers can avoid traditional checkout lines. The system senses when the customer leaves the store, totals the bill, and charges it to that customer's Amazon account. Cashiers are replaced by hundreds of cameras that track purchases (Wingfield, 2018).

### 15.6.3.2 *Impulse buying*

Remember our discussion of the dilemma of stocking shelves to promote impulse buying and the difficulties it posed to helping shoppers find things in the store? We can brainstorm about how the SmartKart display can make shoppers aware of items on special as they push the cart past them on the shelves, popping up special offers and alerting customers to discounts, sale items, promotions, additional coupons, and impulse buying opportunities.

By this same token, the SmartKart display can suggest buying additional items or accessories related to items on the list and expanding the shopping list accordingly. For example, if the shopper is buying a vacuum cleaner, this feature will serve as a reminder: "Don't forget to get some extra vacuum cleaner bags." Not only is this a good marketing idea, but it can be a helpful reminder to shoppers, as long as it is not overdone.

### Exercise 15.3 Pursue This SmartKart Design Idea Further

This is definitely a group assignment. Engage your team in ideation, sketching, critiquing, designing, prototyping, and evaluation to come up with further innovative ideas for the shopping ecology.

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## 15.7 SEGUE

In this chapter, we discussed designing the ecology. In the next chapter, we get into designing the interaction for each device in the ecology.